AP® BIOLOGY · UNIT 1

Chemistry of Life

The Comprehensive Study Guide
Water, Elements, and the Four Biological Macromolecules

Clear lessons for all six Unit 1 topics
Annotated diagrams · Key-term glossary · Misconception alerts
58 original exam-style multiple-choice questions
Free-response practice with scoring guidelines
Full answer explanations · One-page review sheet

The Winning Mind

AP® Biology Unit 1: Chemistry of Life — The Comprehensive Study Guide

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First edition

Contents

How to Use This Guide 4

Unit 1 at a Glance 5

PART 1 · THE LESSONS

Topic 1.1 Structure of Water and Hydrogen Bonding 6

Topic 1.2 Elements of Life 11

Topic 1.3 Introduction to Biological Macromolecules 15

Topic 1.4 Properties of Biological Macromolecules 19

Topic 1.5 Structure and Function of Biological Macromolecules 26

Topic 1.6 Nucleic Acids 31

PART 2 · SKILLS AND REVIEW

Science Practices and Free-Response Strategy 35

The Four Macromolecules: Master Comparison 37

Unit 1 One-Page Review Sheet 38

PART 3 · PRACTICE TEST

Section I: Multiple Choice 39

Section II: Free Response 44

PART 4 · ANSWERS AND REFERENCE

Answers: Check Your Understanding 48

Answers: Practice Test 50

Glossary 54

Score Tracker 56

How to Use This Guide

Unit 1 is the foundation of the entire AP Biology course. Every later unit—cell membranes, enzymes, cellular respiration, DNA replication, gene expression, evolution—depends on the chemistry you learn here. Students who truly understand *why* water behaves the way it does and *how* structure determines function in macromolecules find the rest of the course far easier.

This guide is organized around the six topics of AP Biology Unit 1. Each topic follows the same pattern so you always know where to look:

- **Learning Targets** — what you should be able to do by the end of the lesson. Use them as a checklist.
- **The Lesson** — a clear explanation of the content, with diagrams and tables.
- **Misconception Alerts** — the mistakes that cost students points year after year.
- **Exam Tips** — how the idea tends to be tested and how to phrase strong answers.
- **Key Terms** — the vocabulary you must be able to use precisely.
- **Check Your Understanding** — exam-style multiple-choice questions. Full explanations are in Part 4.

A Four-Week Study Plan

WEEK	WHAT TO DO
Week 1	Read Topics 1.1 and 1.2. Complete both Check Your Understanding sets. Redo every question you missed, then read the explanation.
Week 2	Read Topics 1.3 and 1.4. Practice drawing dehydration synthesis and hydrolysis from memory until you can do it without looking.
Week 3	Read Topics 1.5 and 1.6. Read the Science Practices chapter and write one practice free-response answer under timed conditions.
Week 4	Study the Master Comparison and the One-Page Review Sheet. Take the full Practice Test under timed conditions, score it, and record your results in the Score Tracker.

Study Habits That Work

- **Retrieve, don't reread.** Close the book and write down everything you remember about a topic, then check. Testing yourself is far more effective than highlighting.
- **Draw it.** If you can sketch a water molecule's hydrogen bonds, an amino acid, and a nucleotide from memory, you understand Unit 1.
- **Explain the "because."** AP questions rarely ask *what*; they ask *why* and *what would happen if*. Practice connecting every fact to its cause.
- **Space it out.** Return to earlier topics every few days. Short, repeated reviews beat one long cram session.

Unit 1 at a Glance

On the AP Biology Exam, Unit 1 has historically made up roughly **8–11% of the multiple-choice score**. Its ideas appear again in free-response questions about membranes, enzymes, genetics, and evolution, so its true weight is larger than that number suggests.

TOPIC	TITLE	THE BIG QUESTION
1.1	Structure of Water and Hydrogen Bonding	How does the polarity of water produce the properties that make life possible?
1.2	Elements of Life	Which elements do living things need, and which macromolecules contain them?
1.3	Introduction to Biological Macromolecules	How are polymers built from monomers, and how are they broken down?
1.4	Properties of Biological Macromolecules	What are the structural components of carbohydrates, lipids, proteins, and nucleic acids?
1.5	Structure and Function of Biological Macromolecules	How does a change in a molecule's structure change its function?
1.6	Nucleic Acids	How do the structures of DNA and RNA differ, and how is DNA built?

THE ONE BIG IDEA

Structure determines function. If you remember only one sentence from Unit 1, make it this one. The shape of water explains its properties. The sequence of amino acids determines a protein's shape, and that shape determines what the protein does. The type of bond between sugars determines whether a polysaccharide is used for energy storage or for structure. On almost every Unit 1 free-response question, the winning answer connects a *specific structural feature* to a *specific function*.

Connections to Other Units

UNIT 1 IDEA	WHERE YOU WILL SEE IT AGAIN
Polarity and hydrophobic interactions	Unit 2 — Why phospholipids form bilayers; why some molecules cross membranes easily and others need channels.
Protein shape	Unit 3 — Enzyme active sites and denaturation. Unit 4 — Receptor proteins in cell signaling.
DNA structure, 5′→3′ direction, base pairing	Unit 6 — Replication, transcription, translation, mutations.
Changes in amino acid sequence	Units 6 and 7 — Mutations, variation, and natural selection.

PART 1 · THE LESSONS

1.1

TOPIC 1.1

Structure of Water and Hydrogen Bonding

LEARNING TARGETS

- Explain how the structure of a water molecule makes it polar.
- Describe how hydrogen bonds form between water molecules and between water and other polar substances.
- Explain how hydrogen bonding produces cohesion, adhesion, surface tension, high specific heat, high heat of vaporization, and the lower density of ice.
- Explain why water is an excellent solvent for polar and ionic substances but not for nonpolar ones.

Water Is a Polar Molecule

A water molecule (H_2O) consists of one oxygen atom covalently bonded to two hydrogen atoms. These are **polar covalent bonds**: the electrons are shared, but not equally. Oxygen is much more **electronegative** than hydrogen, meaning it pulls shared electrons more strongly toward itself.

As a result, the oxygen end of the molecule carries a **partial negative charge (δ−)** and each hydrogen carries a **partial positive charge (δ+)**. Because the molecule is **bent** (about 104.5° between the two O–H bonds) rather than straight, these charges do not cancel. The whole molecule has a negative side and a positive side. A molecule with an uneven distribution of charge like this is called **polar**.

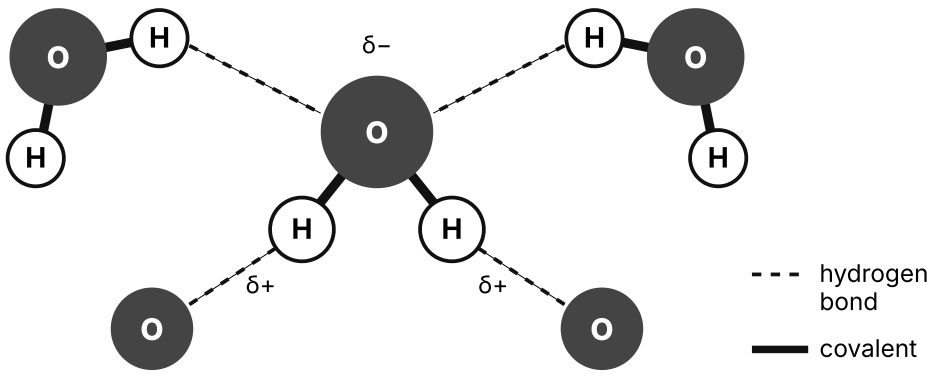


Figure 1.1 Hydrogen bonding in water. The partially negative oxygen of one molecule is attracted to the partially positive hydrogen of a neighbor. Each water molecule can form up to four hydrogen bonds: two through its hydrogens and two through its oxygen.

Hydrogen Bonds

A **hydrogen bond** is a weak attraction between a partially positive hydrogen atom (already covalently bonded to an electronegative atom such as oxygen or nitrogen) and a partially negative atom on another molecule, or on a different part of the same molecule. In liquid water, hydrogen bonds constantly form, break, and re-form.

Any single hydrogen bond is weak—much weaker than a covalent bond. But because there are enormous numbers of them, together they give water unusual and biologically essential properties. Hydrogen bonds also hold together the two strands of DNA and help stabilize the folded shapes of proteins, so you will meet them again in Topics 1.5 and 1.6.

> ◆ **MISCONCEPTION ALERT**
>
> **Hydrogen bonds are *between* water molecules, not *inside* them.** The bonds holding O and H together within one water molecule are polar covalent bonds. A hydrogen bond is the attraction from one molecule to the next. On the exam, labeling the O–H bond inside a molecule as a "hydrogen bond" is a classic wrong answer.

Emergent Properties of Water

Each property below is a direct consequence of polarity and hydrogen bonding. For every one, learn the property, the *reason*, and an *example in a living system*.

1. Cohesion and Surface Tension

Cohesion is the attraction of water molecules to *each other* through hydrogen bonds. At the surface of a body of water, molecules are hydrogen bonded to neighbors beside and below them but not above, so they pull together tightly and create **surface tension**—a surface that resists being broken.

Biological examples: Cohesion helps hold a continuous column of water together as it is pulled up through the xylem of a tall tree during transpiration. Surface tension allows small organisms such as water striders to walk on a pond surface.

2. Adhesion and Capillary Action

Adhesion is the attraction of water molecules to *other polar or charged surfaces*. When water adheres to the walls of a narrow tube while cohesion pulls more water along behind it, the water climbs upward against gravity. This is **capillary action**.

Biological examples: Water adheres to the cellulose walls of xylem vessels, which helps counteract gravity as water moves up a plant. Adhesion and cohesion working together are why water can move from roots to leaves.

3. High Specific Heat

Specific heat is the amount of energy needed to raise the temperature of 1 gram of a substance by 1°C. Water's is very high (about 4.18 J/g·°C). When water is heated, much of the energy goes into *breaking hydrogen bonds* before the molecules can move faster, so the temperature rises slowly. When water cools, hydrogen bonds form and release heat, so the temperature falls slowly.

Biological examples: Large bodies of water stay at a fairly stable temperature, giving aquatic organisms a stable environment and moderating coastal climates. Because organisms are mostly water, they resist sudden changes in internal temperature.

4. High Heat of Vaporization

For a water molecule to evaporate, its hydrogen bonds to neighbors must be broken, which takes a lot of energy. Water therefore has a high **heat of vaporization**. When water evaporates from a surface, the molecules that leave carry a large amount of heat energy away with them. This is **evaporative cooling**.

Biological examples: Sweating in mammals and transpiration from leaves both cool the organism as water evaporates.

5. Ice Is Less Dense Than Liquid Water

Most substances are densest as solids. Water is unusual. As water freezes, each molecule becomes locked into a stable crystal lattice in which it is hydrogen bonded to four neighbors. This lattice holds the molecules *farther apart* than in liquid water, so ice is less dense and floats.

Biological example: A layer of floating ice insulates the liquid water below it, so lakes and ponds freeze from the top down rather than freezing solid. Aquatic organisms can survive the winter in the liquid water beneath the ice.

6. Water Is an Excellent Solvent

Water is often called the "universal solvent," although it does not dissolve everything. Water dissolves **polar** and **ionic** substances well. When a salt such as NaCl enters water, the $\delta-$ oxygens surround each Na^+ ion and the $\delta+$ hydrogens surround each Cl^- ion, forming **hydration shells** that pull the ions apart and keep them in solution. Polar molecules such as glucose dissolve because they can form hydrogen bonds with water.

- **Hydrophilic** ("water-loving") substances are polar or charged and interact with water. Examples: sugars, ions, many amino acids.
- **Hydrophobic** ("water-fearing") substances are nonpolar and do not interact favorably with water. Examples: oils, fats, the fatty acid tails of phospholipids.

Hydrophobic molecules cluster together in water—not because they strongly attract each other, but because water molecules hydrogen bond to one another and exclude them. These **hydrophobic interactions** drive the formation of cell membranes and the folding of proteins.

> **EXAM TIP**
>
> A complete explanation of any water property has three links: **(1)** water is polar → **(2)** so it forms hydrogen bonds → **(3)** which causes the property. For example: "Water has a high specific heat *because* energy must be used to break hydrogen bonds between polar water molecules before their kinetic energy, and therefore temperature, can increase." Answers that state the property without the hydrogen-bond link usually do not earn the explanation point.

Acids, Bases, and pH (Background)

A small fraction of water molecules dissociate into hydrogen ions (H^+) and hydroxide ions (OH^-). The **pH scale** (0–14) measures the concentration of H^+: $pH = -\log[H^+]$. Acids increase $[H^+]$ and lower the pH; bases decrease $[H^+]$ and raise the pH. Each one-unit change in pH is a **tenfold** change in $[H^+]$, so a solution at pH 4 has 100 times more H^+ than one at pH 6. **Buffers** resist changes in pH and help keep conditions stable in cells and blood (pH about 7.4). pH matters in Unit 1 because changes in pH can disrupt the hydrogen and ionic bonds that hold proteins in their functional shape.

KEY TERMS

Polar covalent bond
A covalent bond in which electrons are shared unequally.

Electronegativity
An atom's attraction for shared electrons.

Hydrogen bond
A weak attraction between a δ+ hydrogen and a δ− atom (often O or N) of another molecule or region.

Cohesion
Attraction between molecules of the same substance.

Adhesion
Attraction between molecules of different substances.

Surface tension
Resistance of a liquid's surface to being stretched or broken.

Capillary action
Movement of water up a narrow space due to adhesion and cohesion.

Specific heat
Energy needed to raise 1 g of a substance by 1°C.

Heat of vaporization
Energy needed to convert a liquid to a gas.

Hydrophilic
Polar or charged; dissolves in or interacts with water.

Hydrophobic
Nonpolar; does not interact favorably with water.

Solvent / solute
The dissolving agent / the substance dissolved.

Check Your Understanding 1.1

1. Which of the following best explains why a water molecule is polar?

(A) Hydrogen is more electronegative than oxygen, and the molecule is linear.

(B) Oxygen is more electronegative than hydrogen, and the molecule is bent, so the partial charges do not cancel.

(C) Water contains ionic bonds between oxygen and hydrogen.

(D) Water molecules form hydrogen bonds with one another.

2. A student places a narrow glass tube into a beaker of water and observes the water rise inside the tube above the level in the beaker. Which pair of properties is most directly responsible?

(A) High specific heat and high heat of vaporization

(B) Cohesion and adhesion

(C) Low density of ice and surface tension

(D) Hydrophobic interactions and ionic bonding

3. Coastal cities typically have smaller daily and seasonal temperature swings than inland cities at the same latitude. Which property of water best accounts for this?

(A) Water has a high specific heat because energy is absorbed in breaking hydrogen bonds.

(B) Water is a good solvent for ionic compounds.

(C) Ice floats because it is less dense than liquid water.

(D) Water molecules are held together by covalent bonds.

4. If water molecules were nonpolar, which of the following would most likely be true?

(A) Water would dissolve oils more poorly than it does now.

(B) Water would have a higher boiling point.

(C) Sweating would be a less effective way to cool the body.

(D) Ice would become denser and sink more slowly.

5. Which of the following substances would be least likely to dissolve in water?

(A) Sodium chloride (NaCl)

(B) Glucose ($C_6H_{12}O_6$)

(C) A long hydrocarbon chain such as the tail of a fatty acid

(D) An amino acid with a charged R group

6. Which statement correctly explains why a pond freezes from the top down?

(A) Cold water rises because hydrogen bonds break as temperature drops.

(B) As water freezes, hydrogen bonds hold molecules in a lattice that spaces them farther apart, making ice less dense than liquid water.

(C) Ice has a higher specific heat than liquid water.

(D) Ice forms covalent bonds with the air above it.

1.2

TOPIC 1.2

Elements of Life

LEARNING TARGETS

- Describe how atoms and molecules from the environment are needed to build new molecules.
- Identify which macromolecules contain carbon, hydrogen, oxygen, nitrogen, phosphorus, and sulfur.
- Explain why carbon is the backbone of biological molecules.
- Connect the cycling of matter to the flow of energy in living systems.

Living Things Are Built from Recycled Atoms

Organisms do not create matter. Every atom in your body was obtained from your environment—from the food you eat, the water you drink, and the air you breathe. Organisms must constantly take in atoms and small molecules and use them to **build new molecules** for growth, repair, and reproduction. When organisms die or excrete wastes, decomposers return those atoms to the environment, where they can be used again. This is why matter is said to **cycle** through ecosystems.

Energy, in contrast, **flows** in one direction. Organisms capture energy (from sunlight or from chemical bonds), use it to build and maintain complex molecules, and ultimately lose it as heat. Building large, ordered molecules from small ones requires an input of energy; breaking them down can release energy. You will study energy in depth in Unit 3, but the principle starts here: *living systems need a constant supply of both matter and energy*.

The Six Key Elements: CHNOPS

About 96% of the mass of most organisms comes from just four elements: **carbon, hydrogen, oxygen, and nitrogen**. Add **phosphorus** and **sulfur** and you have the six elements that make up the bulk of biological macromolecules. A handy memory device is **CHNOPS**.

ELEMENT	FOUND IN	WHY IT MATTERS / HOW ORGANISMS OBTAIN IT
Carbon (C)	All four macromolecules: carbohydrates, lipids, proteins, nucleic acids	Forms the backbone of all organic molecules. Producers fix carbon from CO_2 in photosynthesis; consumers obtain it by eating.
Hydrogen (H)	All four macromolecules	Part of nearly every organic molecule and of water. Obtained from water and food.
Oxygen (O)	All four macromolecules	Present in hydroxyl, carbonyl, carboxyl, and phosphate groups. Obtained from water, CO_2, O_2, and food.
Nitrogen (N)	**Proteins** (amino group of every amino acid) and **nucleic acids** (nitrogenous bases)	Needed for amino acids and nucleotides. Plants absorb it from soil as nitrate (NO_3^-) or ammonium (NH_4^+); animals get it from food.

ELEMENT	FOUND IN	WHY IT MATTERS / HOW ORGANISMS OBTAIN IT
Phosphorus (P)	**Nucleic acids** (phosphate group of each nucleotide), **phospholipids**, and ATP	Forms the sugar–phosphate backbone of DNA and RNA and the hydrophilic head of phospholipids. Plants absorb phosphate (PO_4^{3-}) from soil.
Sulfur (S)	**Proteins** (in the amino acids cysteine and methionine)	Sulfur atoms in cysteine form **disulfide bridges** that stabilize protein shape. Plants absorb sulfate (SO_4^{2-}) from soil.

MUST-KNOW ELEMENT RULES

- **Carbohydrates** contain C, H, and O, usually in a 1 : 2 : 1 ratio $(CH_2O)_n$. No N, P, or S in simple sugars.
- **Lipids** are mostly C and H with a little O. **Phospholipids** also contain P (and often N).
- **Proteins** always contain C, H, O, and N. Many contain S.
- **Nucleic acids** always contain C, H, O, N, and P.

A common exam setup: *"A researcher grows cells with radioactive ^{35}S (or ^{32}P or ^{15}N). Which molecules will become labeled?"* Sulfur labels proteins; phosphorus labels nucleic acids (and phospholipids and ATP); nitrogen labels both proteins and nucleic acids.

Why Carbon?

Carbon has **four valence electrons**, so it can form **four covalent bonds**. This lets carbon atoms bond to each other and to many other elements, producing chains, branches, and rings of nearly limitless variety. Carbon can also form single, double, and even triple bonds. This versatility is why carbon forms the **skeleton** of every biological macromolecule.

Attached to these carbon skeletons are **functional groups**—small clusters of atoms that give molecules their characteristic chemical behavior. You do not need to memorize every functional group for the AP exam, but recognizing the common ones will help you read molecular diagrams.

GROUP	STRUCTURE	PROPERTY	FOUND IN
Hydroxyl	–OH	Polar	Sugars, alcohols, the 3′ end of nucleic acids
Carboxyl	–COOH	Acidic (releases H^+)	Amino acids, fatty acids
Amino	$-NH_2$	Basic (accepts H^+)	Amino acids
Phosphate	$-OPO_3^{2-}$	Acidic, negatively charged	Nucleotides, ATP, phospholipids
Carbonyl	C=O	Polar	Sugars
Sulfhydryl	–SH	Forms disulfide bridges	Cysteine
Methyl	$-CH_3$	Nonpolar	Many molecules; DNA methylation

◆ **MISCONCEPTION ALERT**

"Plants get their mass from the soil." They don't—most of a plant's dry mass comes from carbon fixed out of **CO_2 in the air**. Soil supplies smaller amounts of essential elements such as nitrogen, phosphorus, and sulfur. Likewise, the carbon atoms in your body came from the food you ate, which originally came from CO_2 fixed by producers.

● **EXAM TIP**

If a question describes an organism that cannot obtain enough of an element, trace which molecules it cannot build. A plant in nitrogen-poor soil cannot make enough amino acids and nucleotides, so protein synthesis and DNA replication (and therefore growth) slow down. A plant short on phosphorus cannot make enough nucleic acids, ATP, or phospholipids for new membranes.

■ **KEY TERMS**

Element
A substance that cannot be broken down into other substances by chemical reactions.

Organic molecule
A molecule built on a carbon skeleton, typically with C–H bonds.

Covalent bond
A bond formed when atoms share electrons.

Functional group
A group of atoms attached to a carbon skeleton that gives a molecule specific properties.

Disulfide bridge
A covalent S–S bond between two cysteine side chains.

Matter cycles, energy flows
Atoms are reused; energy enters, is used, and leaves as heat.

Check Your Understanding 1.2

1. A researcher grows bacteria in a medium containing radioactive sulfur (^{35}S). Which type of macromolecule in the bacteria will contain the greatest amount of radioactive label?

(A) DNA
(B) Proteins
(C) Carbohydrates
(D) Triglycerides

2. Which pair of macromolecules both contain nitrogen as a required component of their monomers?

(A) Carbohydrates and lipids
(B) Lipids and nucleic acids
(C) Carbohydrates and proteins
(D) Proteins and nucleic acids

3. A crop is grown in soil that is severely deficient in phosphate. Which process would be most directly impaired?

(A) Synthesis of DNA and new phospholipid membranes
(B) Synthesis of cellulose for cell walls
(C) Synthesis of starch for storage
(D) Synthesis of triglycerides in seeds

4. Which property of carbon best explains why it forms the backbone of biological macromolecules?

(A) Carbon is the most abundant element in Earth's crust.

(B) Carbon forms ionic bonds easily with metals.

(C) Carbon can form four covalent bonds, allowing chains, branches, and rings.

(D) Carbon is highly electronegative and forms hydrogen bonds.

5. Most of the dry mass of a large oak tree is made of atoms that originally came from

(A) minerals in the soil

(B) water absorbed by the roots

(C) carbon dioxide from the air

(D) nitrogen fixed by soil bacteria

1.3

TOPIC 1.3

Introduction to Biological Macromolecules

LEARNING TARGETS

- Distinguish between monomers and polymers.
- Describe how dehydration synthesis forms covalent bonds between monomers.
- Describe how hydrolysis breaks covalent bonds between monomers.
- Explain the roles of water and enzymes in building and breaking polymers.

Monomers and Polymers

Biological macromolecules are large molecules built from smaller, repeating subunits. The subunits are called **monomers** ("one part"), and the chains they form are called **polymers** ("many parts"). Three of the four classes of macromolecules are true polymers. Lipids are the exception: they are not built from a chain of repeating monomers, although they are assembled by the same type of reaction.

MACROMOLECULE	MONOMER	POLYMER	BOND BETWEEN MONOMERS
Carbohydrates	Monosaccharide	Polysaccharide	Glycosidic linkage
Proteins	Amino acid	Polypeptide	Peptide bond
Nucleic acids	Nucleotide	Polynucleotide (DNA, RNA)	Phosphodiester bond
Lipids	Not true polymers. A triglyceride is built from glycerol + 3 fatty acids.		Ester linkage

Dehydration Synthesis: Building Polymers

Monomers are joined by **covalent bonds** in a reaction called **dehydration synthesis** (also called a condensation reaction). One monomer gives up a hydroxyl group (–OH) and the other gives up a hydrogen atom (–H). These combine to form a **molecule of water**, which is released, and the two monomers become covalently bonded.

- "De-hydration" = removing water. "Synthesis" = building.
- For each new bond formed, **one water molecule is released**. Building a chain of n monomers releases $n - 1$ water molecules.
- Dehydration synthesis **requires energy** and is catalyzed by enzymes.

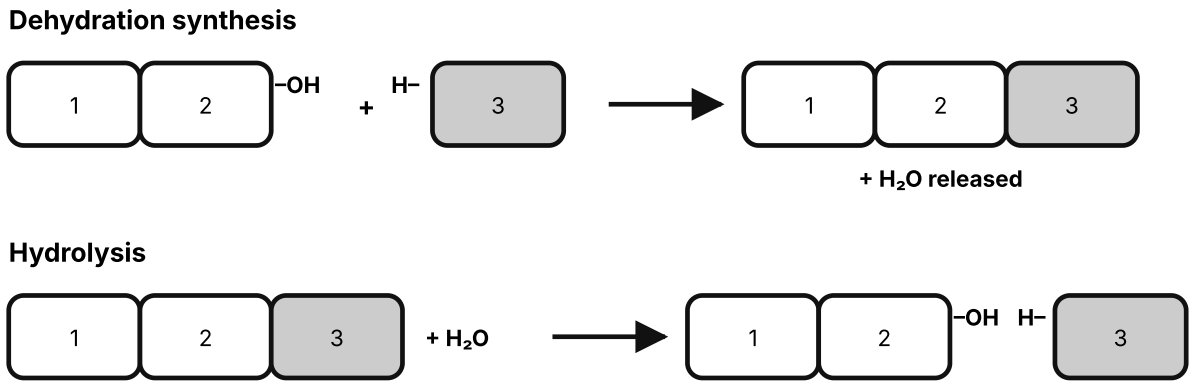


Figure 1.3 Building and breaking polymers. In dehydration synthesis, an –OH from one monomer and an –H from another are removed as water, forming a covalent bond. Hydrolysis is the reverse: adding water breaks the bond, and the –OH and –H are restored to the separated monomers.

Hydrolysis: Breaking Polymers

Hydrolysis ("water-splitting") is the reverse of dehydration synthesis. A **water molecule is added** across the covalent bond between two monomers. The water splits: an –H attaches to one monomer and an –OH attaches to the other, and the bond breaks.

- Digestion is hydrolysis. Enzymes in your digestive system hydrolyze starch into glucose, proteins into amino acids, and fats into glycerol and fatty acids, so the monomers can be absorbed and reused.
- Hydrolysis of polymers generally **releases energy**. For example, the hydrolysis of ATP releases energy that cells use to do work.

SIDE-BY-SIDE SUMMARY

	DEHYDRATION SYNTHESIS	HYDROLYSIS
Result	Builds polymers from monomers	Breaks polymers into monomers
Water	Released (one per bond formed)	Consumed (one per bond broken)
Energy	Requires energy input	Releases energy
Example	Plant cells linking glucose into starch	Digestive enzymes breaking starch into glucose

QUICK MATH

A polypeptide of 150 amino acids has **149 peptide bonds**, and its synthesis released **149 water molecules**. Completely hydrolyzing it requires **149 water molecules**. Rule: n monomers → $n - 1$ bonds → $n - 1$ waters. (For a triglyceride, glycerol + 3 fatty acids forms 3 ester bonds and releases 3 waters.)

The Role of Enzymes

Both dehydration synthesis and hydrolysis are catalyzed by **enzymes**, proteins that speed up chemical reactions without being consumed. Without enzymes, these reactions would happen far too slowly to sustain life. Each enzyme is specific to a particular type of bond, which is why humans can hydrolyze starch but not cellulose (see Topic 1.5).

◆ **MISCONCEPTION ALERT**

Mixing up the names. Students often reverse these terms. Anchor them to the word roots: *hydro* = water, *lysis* = splitting. Hydrolysis uses water to split. Dehydration means "losing water," and that water is lost *as* the bond is made.

■ **KEY TERMS**

Monomer
A small subunit that serves as a building block of a polymer.

Polymer
A long molecule made of many similar or identical monomers linked by covalent bonds.

Macromolecule
A very large molecule, such as a polysaccharide, protein, or nucleic acid.

Dehydration synthesis
A reaction that joins two molecules by removing a water molecule.

Hydrolysis
A reaction that breaks a bond by adding a water molecule.

Enzyme
A biological catalyst, usually a protein, that speeds up reactions.

Check Your Understanding 1.3

1. How many water molecules are released when a polysaccharide of 40 glucose monomers is synthesized in a single unbranched chain?

(A) 20
(B) 40
(C) 39
(D) 41

2. Which of the following correctly describes hydrolysis?

(A) Water is removed to form a covalent bond between monomers.
(B) Water is added to break a covalent bond between monomers.
(C) Hydrogen bonds between water molecules are broken by heat.
(D) Monomers are joined by ionic bonds when water is present.

3. During digestion, proteins in food are broken into amino acids, which are absorbed and later used to build new proteins in the body. Which sequence of reactions correctly describes this process?

(A) Hydrolysis, then dehydration synthesis
(B) Dehydration synthesis, then hydrolysis
(C) Hydrolysis only
(D) Dehydration synthesis only

4. Which macromolecule class is NOT composed of a chain of repeating monomers?

(A) Carbohydrates

(B) Proteins

(C) Nucleic acids

(D) Lipids

5. A scientist adds an enzyme to a solution of maltose (a disaccharide of two glucose molecules) and measures the products. Which observation would indicate that hydrolysis occurred?

(A) The amount of water in the solution decreases and glucose appears.

(B) The amount of water in the solution increases and glucose appears.

(C) The amount of maltose increases and water is released.

(D) Starch appears in the solution.

1.4

TOPIC 1.4

Properties of Biological Macromolecules

LEARNING TARGETS

- Describe the structure of the monomers of carbohydrates, proteins, and nucleic acids, and the components of lipids.
- Explain how the R group of an amino acid determines its chemical properties (nonpolar, polar, or charged).
- Explain the directionality of nucleic acids (5′ and 3′ ends) and of polypeptides (N-terminus and C-terminus).
- Distinguish saturated from unsaturated fatty acids and describe the structure of phospholipids.

In this topic you will learn the parts list for each macromolecule. In Topic 1.5 you will see how those parts determine what each molecule does. Focus on being able to **recognize and draw** each monomer and to identify which parts are polar and which are nonpolar.

Carbohydrates

Carbohydrates include sugars and the polymers made from them. They are used for **energy** (glucose, starch, glycogen) and for **structure** (cellulose, chitin). They are made of C, H, and O.

Monosaccharides

Monosaccharides (simple sugars) are the monomers of carbohydrates. Most have the formula $(CH_2O)_n$, with several **hydroxyl (–OH) groups** that make them polar and water-soluble. Glucose, fructose, and galactose all have the formula $C_6H_{12}O_6$ but differ in the arrangement of their atoms, so they are **isomers** with different properties. Ribose and deoxyribose are five-carbon sugars found in nucleic acids. In water, most five- and six-carbon sugars form **rings**, and the carbons are numbered 1′, 2′, 3′, and so on.

Disaccharides and Polysaccharides

Two monosaccharides joined by dehydration synthesis form a **disaccharide**; the covalent bond is a **glycosidic linkage**. Examples: **sucrose** (glucose + fructose, table sugar), **lactose** (glucose + galactose, milk sugar), and **maltose** (glucose + glucose). Long chains of monosaccharides are **polysaccharides**.

POLYSACCHARIDE	ORGANISM	FUNCTION	STRUCTURE
Starch	Plants	Energy storage	α-glucose monomers; unbranched (amylose) or branched (amylopectin); helical
Glycogen	Animals, fungi	Energy storage (liver, muscle)	α-glucose; very highly branched, allowing rapid release of glucose
Cellulose	Plants	Structure (cell walls)	β-glucose; straight, unbranched chains that hydrogen bond to neighboring chains, forming strong microfibrils

POLYSACCHARIDE	ORGANISM	FUNCTION	STRUCTURE
Chitin	Fungi, arthropods	Structure (cell walls, exoskeletons)	β-linked glucose derivative containing nitrogen

Directionality: Because glycosidic linkages connect specific numbered carbons (for example, carbon 1 of one glucose to carbon 4 of the next, a "1—4 linkage"), polysaccharide chains have two distinct ends.

Lipids

Lipids are a diverse group defined by one shared property: they are mostly **nonpolar and hydrophobic**, because they consist mostly of hydrocarbon regions (C–H bonds share electrons nearly equally). Lipids are not polymers. The three most important groups are fats, phospholipids, and steroids.

Fats (Triglycerides)

A **triglyceride** is made of one **glycerol** molecule (a three-carbon alcohol) joined to **three fatty acids** by **ester linkages** formed in dehydration synthesis. Each **fatty acid** is a long hydrocarbon chain with a carboxyl group (–COOH) at one end. Fats store about twice as much energy per gram as carbohydrates, and they also provide insulation and cushioning.

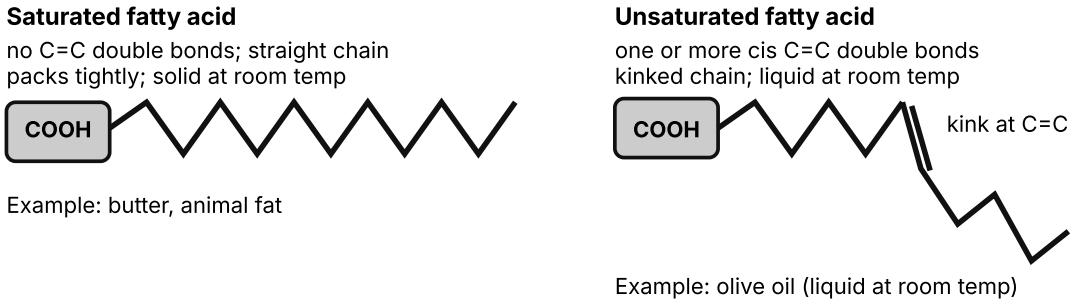


Figure 1.4a Saturated vs. unsaturated fatty acids. A cis double bond creates a bend that keeps neighboring molecules from packing tightly together.

- **Saturated fatty acids** have **no carbon–carbon double bonds**; every carbon holds as many hydrogens as possible ("saturated with hydrogen"). The chains are straight and pack tightly, so saturated fats (most animal fats) are **solid at room temperature**.
- **Unsaturated fatty acids** have **one or more double bonds**. A *cis* double bond causes a **kink** in the chain that prevents tight packing, so unsaturated fats (most plant and fish oils) are **liquid at room temperature**.

Phospholipids

A **phospholipid** has **glycerol** attached to **two fatty acids** and a **phosphate group** (often linked to another small charged or polar group). This gives the molecule a dual personality: it is **amphipathic**, meaning it has both a hydrophilic region and a hydrophobic region.

- The **phosphate "head"** is charged/polar and **hydrophilic**.
- The two **fatty acid "tails"** are nonpolar and **hydrophobic**.

In water, phospholipids spontaneously arrange into a **bilayer**: heads face the watery environment on both sides, and tails face inward, shielded from water. This bilayer is the basic structure of every cell membrane (Unit 2).

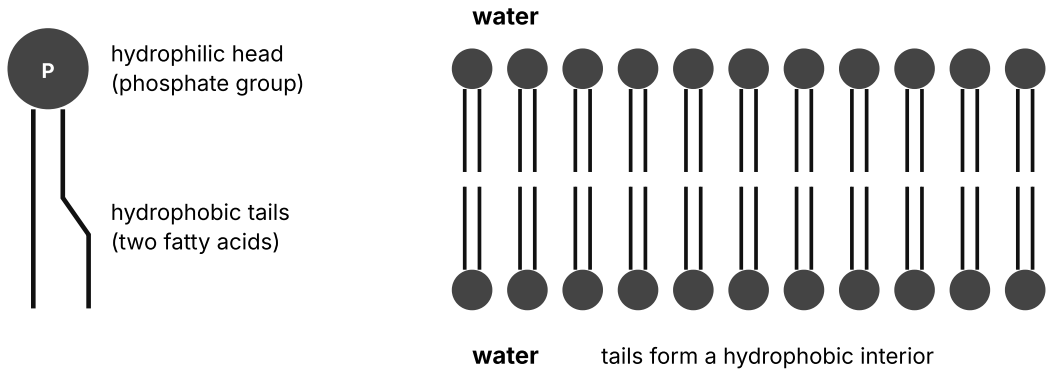


Figure 1.4b A phospholipid (left) and a phospholipid bilayer (right). The hydrophilic heads face water on both sides; the hydrophobic tails face each other in the interior.

Steroids

Steroids are lipids built from **four fused carbon rings**. **Cholesterol** is a component of animal cell membranes, where it helps regulate fluidity, and it is the starting material for steroid hormones such as estrogen and testosterone. Because steroids are nonpolar, steroid hormones can diffuse directly through the cell membrane.

Proteins

Proteins are the most functionally diverse macromolecules. They act as enzymes, structural supports, transport channels, receptors, hormones, antibodies, and motor proteins. Proteins are polymers of **amino acids**.

Amino Acid Structure

Every amino acid has a central carbon (the α carbon) bonded to four things:

1. an **amino group** ($-NH_2$),
2. a **carboxyl group** ($-COOH$),
3. a **hydrogen atom**, and
4. a variable side chain called the **R group**.

There are **20 amino acids** commonly found in proteins. The first three parts are identical in all of them; **only the R group differs**. The R group gives each amino acid its unique chemical properties.

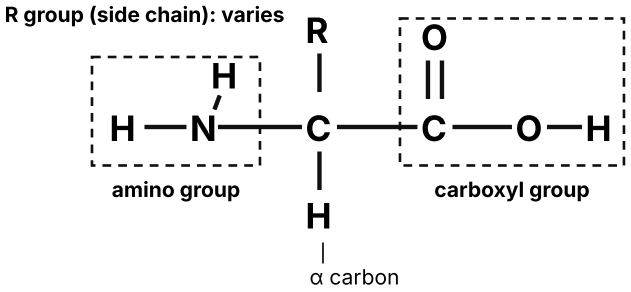


Figure 1.4c General structure of an amino acid. Only the R group differs among the 20 amino acids.

Classifying R Groups

R GROUP TYPE	DESCRIPTION	WHERE IT TENDS TO END UP IN A FOLDED PROTEIN
Nonpolar (hydrophobic)	Mostly C and H, e.g., valine, leucine, phenylalanine	Buried in the protein's interior, away from water; or embedded in the hydrophobic core of a membrane
Polar (hydrophilic)	Contain –OH, –SH, or amide groups, e.g., serine, threonine	On the surface, forming hydrogen bonds with water
Charged: acidic	Negative charge at cell pH (carboxyl in R group), e.g., aspartic acid, glutamic acid	On the surface; can form ionic bonds with positive R groups
Charged: basic	Positive charge at cell pH (amino in R group), e.g., lysine, arginine	On the surface; can form ionic bonds with negative R groups

Peptide Bonds and Directionality

Amino acids are joined by dehydration synthesis between the **carboxyl group of one amino acid** and the **amino group of the next**. The resulting covalent bond is a **peptide bond**. A chain of amino acids is a **polypeptide**. Every polypeptide has **directionality**: one end has a free amino group (the **N-terminus**) and the other has a free carboxyl group (the **C-terminus**). Amino acids are always added to the C-terminus end, and the sequence is always written from N-terminus to C-terminus.

Nucleic Acids

Nucleic acids—DNA and RNA—store, transmit, and help express hereditary information. They are polymers of **nucleotides**. Each nucleotide has three parts:

1. a **five-carbon (pentose) sugar**: deoxyribose in DNA, ribose in RNA;
2. a **phosphate group** attached to the 5′ carbon of the sugar; and
3. a **nitrogenous base** attached to the 1′ carbon of the sugar.

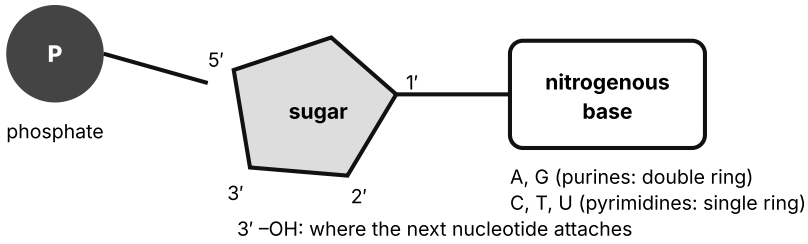


Figure 1.4d A nucleotide. The phosphate is on the 5′ carbon, the base is on the 1′ carbon, and the 3′ carbon carries the hydroxyl group that bonds to the next nucleotide's phosphate.

Directionality: Nucleotides link when the **phosphate on the 5′ carbon** of one nucleotide forms a covalent bond with the **hydroxyl on the 3′ carbon** of the previous nucleotide. This **phosphodiester bond** creates a repeating **sugar–phosphate backbone**, with the bases sticking out to the side. Every strand therefore has a **5′ end** (free phosphate) and a **3′ end** (free hydroxyl). New nucleotides can be added only to the **3′ end**, so nucleic acids are always built in the **5′ → 3′ direction**. Topic 1.6 covers DNA and RNA structure in detail.

◆ **MISCONCEPTION ALERT**

- **"All lipids are polymers."** They are not; lipids are grouped by being hydrophobic, not by having repeating monomers.
- **"Amino acids differ in their amino or carboxyl groups."** Only the R group differs.
- **"Unsaturated fats have no hydrogens."** They have *fewer* hydrogens than the maximum, because of C=C double bonds.

● **EXAM TIP**

When shown a molecular diagram, identify it by its tell-tale features: a ring with many –OH groups is a **sugar**; long C–H chains are a **lipid**; N–C–C=O repeating along a chain is a **polypeptide**; a phosphate–sugar–base unit is a **nucleotide**. Nitrogen with no phosphorus suggests a protein; nitrogen with phosphorus suggests a nucleic acid.

KEY TERMS

Monosaccharide
A simple sugar; the monomer of carbohydrates.

Glycosidic linkage
Covalent bond between two monosaccharides.

Isomers
Molecules with the same formula but different structures.

Triglyceride
Glycerol + three fatty acids joined by ester linkages.

Saturated fatty acid
No C=C double bonds; straight chain.

Unsaturated fatty acid
One or more C=C double bonds; kinked chain.

Amphipathic
Having both hydrophilic and hydrophobic regions.

Steroid
A lipid with four fused carbon rings, e.g., cholesterol.

R group
The variable side chain of an amino acid.

Peptide bond
Covalent bond between the carboxyl group of one amino acid and the amino group of the next.

N-terminus / C-terminus
The amino and carboxyl ends of a polypeptide.

Nucleotide
Pentose sugar + phosphate + nitrogenous base.

Phosphodiester bond
Links the 3′ carbon of one nucleotide to the 5′ phosphate of the next.

Check Your Understanding 1.4

1. Which feature distinguishes one amino acid from another?

(A) The amino group
(B) The carboxyl group
(C) The R group
(D) The type of bond to the α carbon

2. Which statement best explains why most plant oils are liquid at room temperature while butter is solid?

(A) Plant oils contain more glycerol molecules per fatty acid.
(B) Plant oils contain more unsaturated fatty acids, whose kinks prevent tight packing.
(C) Butter contains phospholipids, which form bilayers.
(D) Butter contains more double bonds, which strengthen interactions between molecules.

3. A short strand of DNA has a free phosphate group at one end. That end is called the

(A) 3′ end
(B) 5′ end
(C) N-terminus
(D) C-terminus

4. A protein that spans a cell membrane has a region that passes through the hydrophobic interior of the bilayer. Which amino acids would most likely be found in that region?

(A) Amino acids with positively charged R groups
(B) Amino acids with negatively charged R groups
(C) Amino acids with polar R groups
(D) Amino acids with nonpolar R groups

5. Which characteristic of phospholipids is most responsible for their spontaneous formation of a bilayer in water?

(A) They are amphipathic, with hydrophilic heads and hydrophobic tails.

(B) They are polymers of fatty acid monomers.

(C) They contain four fused carbon rings.

(D) They form peptide bonds with membrane proteins.

6. Which pair of components is joined when a peptide bond forms?

(A) The R groups of two adjacent amino acids

(B) The amino group of one amino acid and the R group of another

(C) The carboxyl group of one amino acid and the amino group of another

(D) The phosphate of one nucleotide and the sugar of another

1.5

TOPIC 1.5

Structure and Function of Biological Macromolecules

LEARNING TARGETS

- Explain how the sequence and chemical properties of monomers determine the structure and function of a polymer.
- Describe the four levels of protein structure and the bonds or interactions that stabilize each.
- Predict how a change in a monomer's sequence or in environmental conditions affects a polymer's structure and function.
- Explain how differences in glycosidic linkages and in fatty acid saturation lead to different functions.

The Central Principle

The function of a macromolecule depends on its three-dimensional **shape**, and its shape depends on the **sequence and chemical properties of its monomers**. Change the monomers, or the order of the monomers, and you may change the shape—and therefore the function. This idea runs through all four classes of macromolecules, but it is most dramatic in proteins.

The Four Levels of Protein Structure

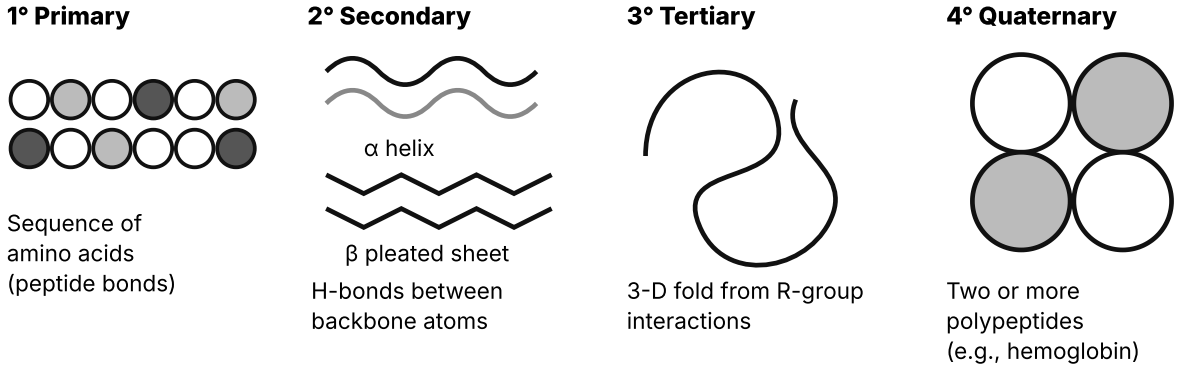


Figure 1.5 The four levels of protein structure. Each level builds on the one before it; ultimately, the primary structure determines all higher levels.

LEVEL	WHAT IT IS	WHAT HOLDS IT TOGETHER
Primary (1°)	The unique **sequence of amino acids** in the polypeptide, determined by the gene that codes for it	**Covalent peptide bonds**
Secondary (2°)	Local, repeating folds: the **α helix** (coil) and the **β pleated sheet**	**Hydrogen bonds between atoms of the polypeptide backbone** (not the R groups)
Tertiary (3°)	The overall **three-dimensional shape** of a single polypeptide	**Interactions between R groups**: hydrophobic interactions (nonpolar R groups cluster inside), hydrogen bonds, ionic bonds, and covalent **disulfide bridges** between cysteines
Quaternary (4°)	Association of **two or more polypeptide subunits** into one functional protein. Only some proteins have it.	The same interactions as tertiary structure, between subunits

◆ **MISCONCEPTION ALERT**

Secondary vs. tertiary bonds. Both involve hydrogen bonds, but in secondary structure they form between *backbone* atoms (the C=O of one amino acid and the N–H of another), whereas in tertiary structure they form between *R groups*. Also, quaternary structure is *not* "just a more folded" protein—it requires more than one polypeptide chain.

When Protein Structure Changes

Changes in Sequence

Because the primary structure determines how the protein folds, even a **single amino acid substitution** can alter the protein's shape and function. The classic example is **sickle-cell disease**. In the β-globin chains of hemoglobin, a single amino acid—glutamic acid, which has a charged, hydrophilic R group—is replaced with valine, which has a nonpolar, hydrophobic R group. The new hydrophobic patch on the surface causes hemoglobin molecules to stick together into long fibers when oxygen levels are low, distorting red blood cells into a rigid sickle shape that can block small blood vessels.

Not every change is harmful. A substitution with an amino acid of similar properties (for example, one nonpolar R group for another), or a change in a region not critical to the protein's shape, may have little or no effect. When predicting the effect of a substitution, ask: *Does the new R group have different chemical properties (size, polarity, charge)? Is the change in a functionally important region, such as an enzyme's active site?*

Changes in Environment: Denaturation

A protein's shape is held together largely by weak interactions (hydrogen bonds, ionic bonds, hydrophobic interactions). Conditions that disrupt these interactions can cause a protein to unfold, or **denature**, and lose its function:

- **High temperature** increases molecular motion enough to break hydrogen bonds and other weak interactions.
- **Changes in pH** alter the charges on acidic and basic R groups, disrupting ionic bonds and hydrogen bonds.
- **High salt concentrations** or certain chemicals interfere with ionic bonds and hydrophobic interactions.

Denaturation does **not** usually break peptide bonds, so the primary structure remains intact. Some proteins can refold if normal conditions return, but many cannot.

> **EXAM TIP**
> A strong "predict and justify" answer about a protein follows this chain: **change in amino acid (or condition) → change in the interactions between R groups → change in 3-D shape → change in function** (for example, the substrate no longer fits the active site). Name the specific interaction that is disrupted whenever you can.

Carbohydrates: The Linkage Determines Function

Starch and cellulose are both polymers of glucose, yet one is food and the other is wood. The difference lies in the type of glucose and the resulting glycosidic linkage.

- **Starch and glycogen** use **α-glucose** with **α(1→4) linkages** (plus α(1→6) branch points). The chains coil into helices and are easily hydrolyzed by enzymes such as amylase, making them good **energy stores**. Glycogen's extensive branching gives it many ends from which glucose can be released quickly.
- **Cellulose** uses **β-glucose** with **β(1→4) linkages**, in which every other glucose is flipped. The chains are straight, and the –OH groups of parallel chains form many **hydrogen bonds** with one another, producing strong, rigid **microfibrils**—ideal for **structural support** in plant cell walls.

Most animals, including humans, lack enzymes that can hydrolyze β(1→4) linkages, so cellulose passes through our digestive tract as fiber. Cows and termites can digest cellulose only because of microorganisms in their guts that produce the necessary enzymes.

Lipids: Saturation Determines Fluidity

The structure of fatty acid tails controls how tightly lipids pack. In membranes, phospholipids with **unsaturated** (kinked) tails cannot pack closely, which keeps the membrane **more fluid**, especially at low temperatures. Phospholipids with **saturated** (straight) tails pack tightly, making the membrane **less fluid**. Organisms adjust membrane composition to their environment: cold-adapted organisms tend to have a higher proportion of unsaturated fatty acids in their membranes.

Nucleic Acids: Sequence Stores Information

The function of DNA and RNA lies in the **sequence of their nitrogenous bases**. This sequence encodes the information used to build proteins. A change in the base sequence (a mutation) can change the amino acid sequence of the encoded protein, which can change the protein's shape and function. Thus, structure determines function at two levels: the sequence of DNA determines the sequence of a protein, which determines its shape and activity.

KEY TERMS

Primary structure
The amino acid sequence of a polypeptide.

Secondary structure
α helices and β sheets stabilized by backbone hydrogen bonds.

Tertiary structure
Overall 3-D shape from R-group interactions.

Quaternary structure
Two or more polypeptides joined into one protein.

Hydrophobic interaction
Clustering of nonpolar groups away from water.

Denaturation
Loss of a protein's shape (and function) due to environmental change.

α vs. β linkage
Orientation of the glycosidic bond; determines starch vs. cellulose.

Membrane fluidity
How freely membrane components move; increased by unsaturated tails.

Check Your Understanding 1.5

1. Which level of protein structure is directly stabilized by hydrogen bonds between the carbonyl oxygen of one amino acid and the amino hydrogen of another amino acid in the backbone?

(A) Primary
(B) Secondary
(C) Tertiary
(D) Quaternary

2. An enzyme from a human stomach (pH 2) is placed in a solution at pH 8 and loses its activity. The most likely explanation is that

(A) the peptide bonds of the enzyme were hydrolyzed
(B) the enzyme's primary structure was changed
(C) changes in the charges of R groups disrupted ionic and hydrogen bonds, altering the enzyme's shape
(D) the enzyme gained additional polypeptide subunits

3. In a protein, a polar amino acid on the surface is replaced by another polar amino acid of similar size. Which outcome is most likely?

(A) Little or no change in the protein's structure and function
(B) Complete loss of the protein's tertiary structure
(C) Conversion of the protein into a lipid
(D) Loss of all peptide bonds in the protein

4. Cellulose and starch are both polymers of glucose. Why can humans digest starch but not cellulose?

(A) Cellulose contains nitrogen, which humans cannot process.
(B) Starch is a lipid, and cellulose is a carbohydrate.
(C) Cellulose molecules are too small to be absorbed.
(D) Human enzymes can hydrolyze α-glycosidic linkages but not the β-glycosidic linkages in cellulose.

5. A fish species living in near-freezing Arctic water is compared with a related species from tropical waters. Which prediction about the Arctic species' cell membranes is best supported?

(A) A higher proportion of saturated fatty acids, to keep the membrane rigid

(B) A higher proportion of unsaturated fatty acids, to keep the membrane fluid

(C) No phospholipids, to prevent freezing

(D) More peptide bonds in membrane phospholipids

6. Hemoglobin consists of four polypeptide chains. Which level of structure describes the arrangement of these four chains?

(A) Primary

(B) Secondary

(C) Tertiary

(D) Quaternary

1.6

TOPIC 1.6

Nucleic Acids

LEARNING TARGETS

- Describe the structural similarities and differences between DNA and RNA.
- Explain how DNA's antiparallel double helix is held together.
- Apply complementary base-pairing rules (A–T, G–C; A–U in RNA).
- Explain why nucleic acids are synthesized in the 5′ → 3′ direction.

DNA and RNA Compared

FEATURE	DNA (DEOXYRIBONUCLEIC ACID)	RNA (RIBONUCLEIC ACID)
Sugar	**Deoxyribose** — has an H at the 2′ carbon ("de-oxy" = missing an oxygen)	**Ribose** — has an –OH at the 2′ carbon
Bases	Adenine (A), guanine (G), cytosine (C), **thymine (T)**	Adenine (A), guanine (G), cytosine (C), **uracil (U)**
Strands	Usually **double-stranded** (double helix)	Usually **single-stranded**; can fold into complex shapes by base pairing within itself
Function	Long-term storage of genetic information	Carries and helps express information (mRNA, tRNA, rRNA); some RNAs act as catalysts
Stability	More chemically stable	Less stable; the 2′ –OH makes it more reactive

Shared features: Both are polymers of nucleotides; both have a sugar–phosphate backbone joined by phosphodiester bonds; both have 5′ and 3′ ends; both use A, G, and C.

Nitrogenous Bases: Purines and Pyrimidines

- **Purines** have a **double-ring** structure: **adenine (A)** and **guanine (G)**. Memory hook: "Pure As Gold."
- **Pyrimidines** have a **single-ring** structure: **cytosine (C)**, **thymine (T)**, and **uracil (U)**. Memory hook: "CUT the Py."

In the double helix, a purine always pairs with a pyrimidine. A double-ring base paired with a single-ring base gives every rung of the DNA "ladder" the same width, so the helix has a uniform diameter.

The Structure of the Double Helix

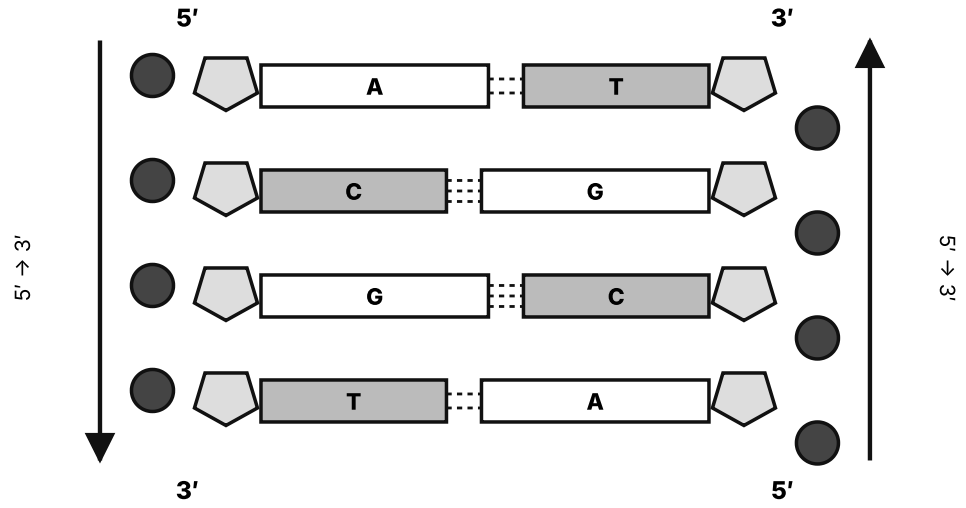


Figure 1.6 DNA's antiparallel structure, shown untwisted. Dark circles are phosphates and gray pentagons are deoxyribose sugars. White bars are purines (A, G); gray bars are pyrimidines (C, T). Dashed lines are hydrogen bonds: two between A and T, three between G and C.

- **Sugar–phosphate backbones** run along the outside of the helix, held together by **covalent phosphodiester bonds**. The negatively charged phosphates face the watery environment.
- **Nitrogenous bases** point inward and pair across the two strands by **hydrogen bonds**.
- **Complementary base pairing:** A pairs with T (**two** hydrogen bonds) and G pairs with C (**three** hydrogen bonds). In RNA, A pairs with U.
- The two strands are **antiparallel**: they run in opposite directions. One strand runs 5′ → 3′ from top to bottom while its partner runs 3′ → 5′.

WHY THESE STRUCTURAL FACTS MATTER

- **Complementary pairing** means each strand can serve as a template for building the other, which is the basis of DNA replication and transcription (Unit 6).
- **Hydrogen bonds** between bases are individually weak, so the strands can be separated ("unzipped") for copying, yet millions of them together keep the molecule stable.
- **G–C pairs have three hydrogen bonds**, so DNA with a higher G–C content requires more energy (a higher temperature) to separate its strands.
- **5′ → 3′ synthesis**: nucleotides can only be added to the free 3′ –OH of a growing strand. This is why the two strands are copied differently during replication.

CHARGAFF'S RULES — QUICK MATH

In double-stranded DNA, the amount of A equals T, and the amount of G equals C. So if a DNA sample is **20% adenine**, then T = 20%, A + T = 40%, leaving 60% for G + C, so **G = 30%** and **C = 30%**. (These rules apply to double-stranded DNA, not to single-stranded RNA.)

Writing a complementary strand: For 5′–ATGCCA–3′, the complementary DNA strand is 3′–TACGGT–5′, which, written 5′ to 3′, is **5′–TGGCAT–3′**. Always label the ends.

◆ **MISCONCEPTION ALERT**

- **"The two strands are held together by covalent bonds."** No—the strands are held together by *hydrogen* bonds between bases. Covalent bonds hold each strand's backbone together.
- **"Antiparallel means the strands are twisted."** Antiparallel refers to *direction* (5′→3′ vs. 3′→5′), not to the helix shape.
- **Forgetting uracil.** RNA uses U instead of T. If a question asks for the RNA complement of A, the answer is U.

■ **KEY TERMS**

Deoxyribose / ribose
The five-carbon sugars of DNA / RNA; ribose has a 2′ –OH.

Purine
Double-ring base: adenine, guanine.

Pyrimidine
Single-ring base: cytosine, thymine, uracil.

Double helix
The two-stranded, twisted structure of DNA.

Antiparallel
Running in opposite 5′→3′ directions.

Complementary base pairing
A–T (or A–U) and G–C pairing by hydrogen bonds.

Sugar–phosphate backbone
Alternating sugars and phosphates linked by phosphodiester bonds.

Chargaff's rules
In dsDNA, %A = %T and %G = %C.

Check Your Understanding 1.6

1. A sample of double-stranded DNA contains 35% guanine. What percentage of the bases are thymine?

(A) 15%
(B) 35%
(C) 30%
(D) 65%

2. Which of the following is found in RNA but not in DNA?

(A) Uracil and a hydroxyl group on the 2′ carbon of the sugar
(B) Thymine and deoxyribose
(C) Phosphodiester bonds and adenine
(D) Hydrogen bonds and guanine

3. What is the complementary DNA sequence to 5′–GATTCA–3′, written in the 5′ to 3′ direction?

(A) 5′–CTAAGT–3′
(B) 5′–ACTTAG–3′
(C) 5′–TGAATC–3′
(D) 5′–UGAAUC–3′

4. Two DNA molecules of equal length are heated. Molecule X has 60% G–C base pairs and molecule Y has 30% G–C base pairs. Which prediction is best supported?

(A) Molecule Y will separate at a higher temperature because A–T pairs have more hydrogen bonds.

(B) Both molecules will separate at the same temperature because they are the same length.

(C) Molecule X will separate at a higher temperature because G–C pairs form three hydrogen bonds.

(D) Neither molecule will separate because the strands are held by covalent bonds.

5. Which statement best describes why the width of the DNA double helix is uniform along its length?

(A) All four bases are the same size.

(B) Each base pair consists of one purine and one pyrimidine.

(C) Purines always pair with purines.

(D) The phosphate groups pair with one another across the helix.

PART 2 · SKILLS AND REVIEW

Science Practices and Free-Response Strategy

The AP Biology Exam has two sections: multiple choice and free response. In recent years the free-response section has included two long questions (one of which involves interpreting an experiment) and four short questions that target specific skills such as experimental design, data analysis, models, and evolutionary trees. Always confirm the current format in the official College Board course description. Whatever the format, the scoring rewards the same habits.

Know Your Task Verbs

Every free-response prompt begins with a task verb. Each verb asks for something different, and answering the wrong verb is the fastest way to lose points.

VERB	WHAT IT ASKS FOR	UNIT 1 EXAMPLE OF A FULL-CREDIT RESPONSE
Identify	Name or state the answer. No explanation needed.	"Identify the bond that forms between amino acids." → A peptide bond.
Describe	Give the relevant characteristics of something.	"Describe the structure of a phospholipid." → A glycerol bonded to two hydrophobic fatty acid tails and a hydrophilic phosphate-containing head.
Explain	Give the *how* or *why*; connect cause and effect.	"Explain why ice floats." → Hydrogen bonds lock frozen water molecules into a lattice that holds them farther apart than in liquid water, so ice is less dense.
Predict	State an expected outcome. A reason is not required unless asked.	"Predict the effect of replacing a hydrophobic amino acid in the core with a charged one." → The protein will misfold and lose function.
Justify	Give evidence or reasoning that supports a claim or prediction.	"Justify your prediction." → A charged R group would be repelled by the hydrophobic core, disrupting hydrophobic interactions that stabilize tertiary structure.
Calculate	Perform a mathematical calculation and show work.	"Calculate the percentage of cytosine if adenine is 22%." → (100 − 44) ÷ 2 = 28%.
Construct / Represent	Draw a graph, table, or diagram.	Plot data with a labeled x-axis (independent variable), labeled y-axis (dependent variable), units, a title, and a key if needed.
Support / Refute	Use evidence to argue for or against a claim.	Cite specific values from the data, then explain how they support or contradict the claim.

Experimental Design Vocabulary

Independent variable
The factor the experimenter deliberately changes (x-axis).

Dependent variable
The factor that is measured in response (y-axis).

Controlled variables (constants)
Conditions kept the same in all groups so they cannot explain differences in results.

Negative control
A group in which no response is expected; shows what happens without the treatment.

Positive control
A group in which a known response is expected; shows the method can detect an effect.

Null hypothesis
The prediction that the independent variable has no effect.

Error bars
Often $\pm 2\ SE_{\bar{x}}$ (about 95% confidence). If error bars of two groups do not overlap, the difference is likely significant; if they overlap a lot, it is likely not.

Five Rules for Free-Response Success

1. **Answer in order and label your parts.** Readers score each part separately. Write (a), (b), (c) and keep each answer next to its label.
2. **Be specific.** Say "hydrogen bonds between the R groups of polar amino acids," not "bonds in the protein."
3. **Use "because."** For every *explain* or *justify* prompt, make sure your sentence contains a cause and an effect.
4. **Use the data.** When a graph or table is given, quote actual numbers with units.
5. **Don't contradict yourself.** Readers do not award credit for a correct answer buried among incorrect ones. Give one clear answer.

THE CLAIM–EVIDENCE–REASONING TEMPLATE

Claim: State your answer in one sentence. **Evidence:** Point to the specific data or structural feature. **Reasoning:** Explain the biological principle that links the evidence to the claim. Example: "*Claim:* Enzyme activity decreases above 45°C. *Evidence:* Activity falls from 9.1 to 1.2 units between 45°C and 60°C. *Reasoning:* High temperatures break hydrogen bonds and other weak interactions that maintain the enzyme's tertiary structure, so the active site changes shape and the substrate no longer binds."

The Four Macromolecules: Master Comparison

	CARBOHYDRATES	LIPIDS	PROTEINS	NUCLEIC ACIDS
Elements	C, H, O	C, H, O (+ P in phospholipids)	C, H, O, N (+ S)	C, H, O, N, P
Monomer	Monosaccharide	None (glycerol + fatty acids)	Amino acid (20 kinds)	Nucleotide
Polymer	Polysaccharide	Not a polymer	Polypeptide	DNA, RNA
Bond	Glycosidic linkage	Ester linkage	Peptide bond	Phosphodiester bond
Polarity	Polar (many –OH groups)	Mostly nonpolar; phospholipids amphipathic	Depends on R groups	Polar backbone; charged phosphates
Directionality	Ends defined by carbon numbering of linkages	—	N-terminus → C-terminus	5′ → 3′
Main functions	Quick energy; energy storage; structure	Long-term energy storage; membranes; hormones; insulation	Enzymes, structure, transport, signaling, defense, movement	Store, transmit, and express genetic information
Examples	Glucose, sucrose, starch, glycogen, cellulose, chitin	Triglycerides, phospholipids, cholesterol, steroid hormones	Enzymes, hemoglobin, collagen, antibodies, channel proteins	DNA, mRNA, tRNA, rRNA (ATP is a related nucleotide)
Structure–function example	α linkages (starch) = digestible storage; β linkages (cellulose) = rigid structure	Unsaturated kinks = more fluid membranes	Amino acid sequence → fold → active-site shape	Base sequence encodes protein sequence; complementary pairing allows copying

Bonds and Interactions: Strongest to Weakest

BOND OR INTERACTION	WHAT IT IS	WHERE IT MATTERS IN UNIT 1
Covalent (nonpolar or polar)	Shared electrons; strong	Within all molecules; peptide, glycosidic, phosphodiester, ester bonds; disulfide bridges
Ionic	Attraction between oppositely charged ions or groups	Salts; between charged R groups in tertiary structure
Hydrogen bond	Attraction between δ+ H and δ− O or N	Water's properties; protein 2° and 3° structure; DNA base pairs; cellulose microfibrils
Hydrophobic interaction	Clustering of nonpolar groups away from water	Protein folding; membrane bilayer formation

Note: In water, ionic bonds are weakened considerably because water molecules surround and separate ions; that is why salts dissolve.

Unit 1 One-Page Review Sheet

1.1 Water

- O is more electronegative than H → polar covalent bonds; bent shape → **polar molecule**.
- **Hydrogen bonds** form *between* water molecules (up to 4 each).
- Cohesion (water–water) → surface tension; transpiration pull.
- Adhesion (water–other) → capillary action in xylem.
- High specific heat → stable temperatures (energy breaks H-bonds first).
- High heat of vaporization → evaporative cooling (sweat, transpiration).
- Ice less dense (H-bond lattice) → floats, insulates lakes.
- Solvent for polar and ionic substances (hydration shells); hydrophobic substances excluded.

1.2 Elements

- **CHNOPS.** N → proteins + nucleic acids. P → nucleic acids + phospholipids + ATP. S → proteins (cysteine, methionine).
- Carbon: 4 covalent bonds → chains, rings, branches.
- Matter cycles; energy flows (lost as heat).

1.3 Building and Breaking

- **Dehydration synthesis:** –OH + –H removed as H_2O → covalent bond. Builds.
- **Hydrolysis:** add H_2O → break bond. Breaks (digestion).
- n monomers → $n - 1$ bonds → $n - 1$ waters.
- Both catalyzed by enzymes.

1.4 Building Blocks

- **Carbs:** monosaccharides, glycosidic linkages. Starch/glycogen = storage; cellulose/chitin = structure.
- **Lipids:** hydrophobic, not polymers. Triglyceride = glycerol + 3 fatty acids. Saturated = no C=C, straight, solid. Unsaturated = C=C kinks, liquid. Phospholipid = amphipathic → bilayer. Steroid = 4 rings.
- **Proteins:** amino acid = amino group + carboxyl + H + **R group**. R groups nonpolar, polar, or charged. Peptide bonds. N- to C-terminus.
- **Nucleic acids:** nucleotide = 5-C sugar + phosphate + base. Phosphodiester bonds; sugar–phosphate backbone. Built 5′ → 3′.

1.5 Structure → Function

- 1° sequence (peptide bonds) → 2° α/β (backbone H-bonds) → 3° R-group interactions (hydrophobic, H, ionic, disulfide) → 4° multiple chains.
- Amino acid change → R-group interactions → shape → function (sickle cell: Glu → Val).
- Denaturation: heat, pH, salt disrupt weak bonds; peptide bonds intact.
- α vs. β linkage → starch vs. cellulose.
- Unsaturated tails → more fluid membranes.

1.6 Nucleic Acids

- DNA: deoxyribose, A/T/G/C, double-stranded. RNA: ribose (2′ –OH), A/U/G/C, single-stranded.
- Purines A, G (2 rings); pyrimidines C, T, U (1 ring).
- A=T (2 H-bonds); G≡C (3 H-bonds). Antiparallel strands.
- Chargaff: %A = %T, %G = %C (dsDNA).

PART 3 · PRACTICE TEST

Section I: Multiple Choice

Time suggestion: 35 minutes · 25 questions. Choose the one best answer for each question. Mark your answers, then check them in Part 4. Record your score in the Score Tracker at the end of the book.

1. Which type of bond or attraction is responsible for holding two separate water molecules together in liquid water?

(A) Polar covalent bond
(B) Nonpolar covalent bond
(C) Hydrogen bond
(D) Ionic bond

2. Some insects can walk across the surface of a pond without sinking. This ability depends most directly on

(A) surface tension resulting from cohesion between water molecules
(B) adhesion of water to the insect's legs
(C) the high specific heat of water
(D) the lower density of ice compared with liquid water

3. Bacteria are grown in a medium in which the only nitrogen source contains the heavy isotope ^{15}N. In which of the following molecules would ^{15}N be found?

(A) Glucose and glycogen
(B) Triglycerides and cholesterol
(C) Proteins only
(D) Both proteins and nucleic acids

4. How many molecules of water are required to completely hydrolyze a tripeptide into its individual amino acids?

(A) 1
(B) 2
(C) 3
(D) 4

5. Which of the following molecules is amphipathic?

(A) A phospholipid
(B) A triglyceride
(C) Glucose
(D) Cellulose

6. A molecule is found to contain carbon, hydrogen, oxygen, nitrogen, and phosphorus. It includes a five-carbon sugar. This molecule is most likely

(A) an amino acid
(B) a fatty acid
(C) a nucleotide
(D) a monosaccharide

7. Disulfide bridges that help stabilize tertiary structure form between the R groups of which amino acid?

(A) Glycine
(B) Cysteine
(C) Valine
(D) Lysine

8. Chitin is a polysaccharide found in the exoskeletons of insects and the cell walls of fungi. Its primary function is most similar to that of

(A) starch in potatoes
(B) glycogen in muscle
(C) sucrose in sugarcane
(D) cellulose in plant cell walls

9. Glycogen is much more highly branched than amylose (unbranched starch). Which of the following is the most likely advantage of this structure for animals?

(A) Branching provides many ends from which glucose can be released rapidly by enzymes.
(B) Branching makes glycogen impossible to hydrolyze, so energy is stored permanently.
(C) Branching allows glycogen to form a rigid structural support.
(D) Branching makes glycogen hydrophobic, so it can be stored in fat cells.

10. A soluble protein folds in the watery cytoplasm of a cell. Where would amino acids with nonpolar R groups most likely be located?

(A) On the outer surface, in contact with water
(B) Only at the N-terminus
(C) In the interior of the protein, away from water
(D) Only at the C-terminus

11. In the active site of an enzyme, a negatively charged R group normally forms an ionic bond with a positively charged region of the substrate. A mutation replaces this amino acid with one that has a positively charged R group. Which outcome is most likely?

(A) The substrate will bind more tightly to the active site.
(B) The substrate will bind less effectively because like charges repel.
(C) The enzyme will gain a fifth level of protein structure.
(D) The enzyme will be converted into a nucleic acid.

12. Which of the following correctly describes a difference between DNA and RNA?

(A) RNA contains ribose, which has a hydroxyl group on the 2′ carbon; DNA contains deoxyribose, which does not.

(B) DNA contains uracil, while RNA contains thymine.

(C) RNA nucleotides lack phosphate groups.

(D) Only DNA has a sugar–phosphate backbone.

13. A sample of double-stranded DNA is 18% cytosine. What percentage of its bases are adenine?

(A) 18%

(B) 36%

(C) 64%

(D) 32%

14. During the synthesis of a nucleic acid, a new nucleotide is added to which part of the growing strand?

(A) The phosphate group at the 5′ end

(B) The nitrogenous base at the 1′ carbon

(C) The hydroxyl group at the 3′ end

(D) The hydroxyl group at the 2′ carbon

15. The two strands of a DNA double helix are described as antiparallel. This means that

(A) the strands are twisted around each other

(B) the strands run in opposite 5′→3′ directions

(C) each strand contains only purines or only pyrimidines

(D) the strands are joined by covalent bonds between bases

16. On a hot day, a dog pants and a person sweats. Both behaviors cool the body because

(A) water has a low specific heat, so it heats up quickly

(B) ice is less dense than liquid water

(C) water is a good solvent for salts

(D) water has a high heat of vaporization, so evaporating molecules remove a large amount of heat

17. When an egg white is cooked, the protein albumin becomes denatured. Which level of protein structure remains intact?

(A) Primary

(B) Secondary

(C) Tertiary

(D) All levels are destroyed

18. The great strength of cellulose fibers in plant cell walls results mainly from

(A) ionic bonds between glucose and calcium ions

(B) peptide bonds between adjacent chains

(C) hydrogen bonds between hydroxyl groups of parallel, straight chains

(D) hydrophobic interactions between fatty acid tails

19. How many molecules of water are released when one triglyceride is formed?

(A) 1
(B) 3
(C) 4
(D) 6

20. Which statement best explains why saturated fats are typically solid at room temperature?

(A) They contain many C=C double bonds.
(B) They contain phosphate groups that bond to one another.
(C) Their straight hydrocarbon chains pack tightly together.
(D) They are polymers of glucose.

21. Glucose dissolves readily in water, but a fatty acid's long hydrocarbon tail does not. The best explanation is that

(A) glucose has polar hydroxyl groups that form hydrogen bonds with water
(B) glucose is larger than a fatty acid
(C) fatty acids contain nitrogen, which repels water
(D) glucose forms covalent bonds with water molecules

22. A plant is exposed to carbon dioxide containing radioactive ^{14}C. After several weeks, in which molecules could the radioactive carbon be found?

(A) Carbohydrates only
(B) Carbohydrates and lipids only
(C) Proteins and nucleic acids only
(D) Carbohydrates, lipids, proteins, and nucleic acids

23. In sickle-cell disease, a glutamic acid (charged R group) in hemoglobin is replaced by valine (nonpolar R group). Which explanation best describes why this substitution changes the protein's function?

(A) Valine cannot form peptide bonds, so the polypeptide is shortened.
(B) The hydrophobic valine on the protein's surface causes hemoglobin molecules to stick together, changing the shape of red blood cells.
(C) The substitution adds a phosphate group to hemoglobin.
(D) The substitution changes hemoglobin from a protein into a lipid.

24. The table shows the temperature at which half of the DNA in a sample separates into single strands (melting temperature, T_m) for DNA samples with different G–C content.

G–C CONTENT (%)	40	50	60	70
T_m (°C)	87	91	95	99

Based on the data, what is the best prediction of T_m for a DNA sample with 55% G–C content?

(A) 89°C
(B) 91°C
(C) 95°C
(D) 93°C

25. Steroid hormones such as testosterone can diffuse directly across the plasma membrane, but glucose cannot. Which property of steroids best explains this difference?

(A) Steroids are nonpolar lipids that can pass through the hydrophobic interior of the bilayer.

(B) Steroids are polymers of amino acids.

(C) Steroids are charged and attracted to the phosphate heads.

(D) Steroids are smaller than all other molecules in the cell.

Section II: Free Response

Time suggestion: 40 minutes · 3 questions. Write your answers in complete sentences on the lines provided or on separate paper. Label each part. Scoring guidelines appear in Part 4.

Question 1 (Long Response, 9 points)

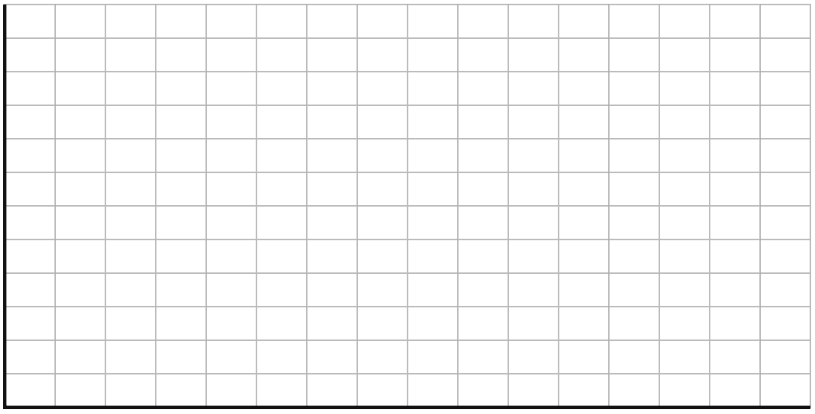
Bacteria cannot regulate their body temperature. To study how they respond to cold, researchers grew genetically identical populations of a bacterial species at four temperatures in the same nutrient medium. After 24 hours, they extracted the membrane phospholipids and measured the percentage of fatty acid tails that were unsaturated. The results are shown below (mean ± 2SE$_{\bar{x}}$; n = 10 cultures per temperature).

GROWTH TEMPERATURE (°C)	10	20	30	40
Unsaturated fatty acids (%)	68 ± 3	57 ± 3	46 ± 2	34 ± 3

(a) Describe the structural difference between a saturated and an unsaturated fatty acid, and **explain** how this difference affects how tightly phospholipids pack in a membrane. **[2 pts]**

(b) Identify the independent variable and the dependent variable in this experiment. **Identify** one variable that was controlled. **[2 pts]**

(c) Construct an appropriately labeled graph of the data, including error bars. **[2 pts]**



(d) Describe the relationship between growth temperature and the percentage of unsaturated fatty acids. Based on the data, **predict** the percentage of unsaturated fatty acids in bacteria grown at 25°C. **[1 pt]**

(e) Explain how the change in membrane composition at low temperature benefits the bacteria. A mutant strain of this species cannot produce unsaturated fatty acids. **Predict** how the growth of the mutant at 10°C would compare with that of the normal strain, and **justify** your prediction. **[2 pts]**

Question 2 (Short Response, 4 points)

Enzyme Z is a soluble protein that functions in the cytoplasm. In its normal folded shape, the amino acid leucine (nonpolar R group) at position 112 is buried in the hydrophobic core of the protein, far from the active site.

(a) Describe two types of interactions between R groups that stabilize the tertiary structure of a protein. **[1 pt]**

(b) A mutation replaces leucine 112 with aspartic acid, which has a negatively charged R group. **Predict** the effect of this mutation on the activity of enzyme Z. **[1 pt]**

(c) Justify your prediction in part (b). **[1 pt]**

(d) A different mutation replaces leucine 112 with isoleucine, which also has a nonpolar R group of similar size. **Explain** why this mutation is likely to have a much smaller effect on enzyme activity than the mutation in part (b). **[1 pt]**

Question 3 (Short Response, 4 points)

A student analyzed the base composition of three nucleic acid samples, each containing thymine. One value is missing.

SAMPLE	%A	%T	%G	%C
1	30	30	20	?
2	24	32	18	26
3	18	18	32	32

(a) Calculate the percentage of cytosine in Sample 1. **[1 pt]**

(b) Identify the sample that is most likely single-stranded DNA and **justify** your answer using the data. **[2 pts]**

(c) Samples 1 and 3 are both double-stranded and the same length. **Explain** which sample would require a higher temperature to separate its two strands. **[1 pt]**

PART 4 · ANSWERS AND REFERENCE

Answers: Check Your Understanding

Topic 1.1

1. (B) Oxygen's greater electronegativity creates partial charges, and the bent shape keeps them from canceling. (A) reverses electronegativity and shape. (C) The bonds are polar covalent, not ionic. (D) Hydrogen bonding is a *result* of polarity, not its cause.

2. (B) Water adheres to the glass and cohesion pulls more water up behind it: capillary action. The other properties do not move water upward.

3. (A) Energy absorbed by water first breaks hydrogen bonds, so water temperature changes slowly and moderates nearby air temperatures.

4. (C) Nonpolar water would not form hydrogen bonds, so far less energy would be needed to evaporate it; evaporation would carry away much less heat. (A) Nonpolar water would dissolve oils *better*. (B) The boiling point would be lower. (D) is not supported.

5. (C) A long hydrocarbon chain is nonpolar and hydrophobic. Ions (A), polar sugars (B), and charged amino acids (D) all interact with water.

6. (B) In ice, stable hydrogen bonds lock molecules into a spacious lattice that is less dense than liquid water, so ice floats and insulates the water below.

Topic 1.2

1. (B) Sulfur is found in the amino acids cysteine and methionine. DNA contains no sulfur; carbohydrates and triglycerides contain no sulfur.

2. (D) Every amino acid has an amino group, and every nucleotide has a nitrogenous base. Simple carbohydrates and triglycerides contain no nitrogen.

3. (A) Phosphorus is required for nucleotides (DNA) and phospholipids (membranes). Cellulose, starch, and triglycerides are made only of C, H, and O.

4. (C) Carbon's four valence electrons allow four covalent bonds, enabling diverse skeletons. (A) is false; (B) and (D) describe the wrong bond types.

5. (C) Most plant dry mass is carbon fixed from atmospheric CO_2 during photosynthesis. Soil minerals and water contribute far less dry mass.

Topic 1.3

1. (C) An unbranched chain of 40 monomers has 39 glycosidic linkages, and each released one water molecule.

2. (B) Hydrolysis adds water to break a covalent bond. (A) describes dehydration synthesis.

3. (A) Digestion breaks food proteins by hydrolysis; cells then build new proteins by dehydration synthesis.

4. (D) Lipids are not built from repeating monomers.

5. (A) Hydrolysis consumes water, so water decreases as glucose (the monomer) is produced.

Topic 1.4

1. (C) All amino acids share the amino group, carboxyl group, and hydrogen; only the R group differs.

2. (B) Cis double bonds in unsaturated fatty acids create kinks that prevent tight packing, so oils stay liquid. (D) reverses the facts: butter is mostly saturated.

3. (B) The 5′ end has a free phosphate on the 5′ carbon. N- and C-terminus describe polypeptides.

4. (D) The interior of the bilayer is hydrophobic, so nonpolar R groups interact favorably with it. Charged or polar R groups would be excluded.

5. (A) Hydrophilic heads face water while hydrophobic tails cluster inside, away from water. Phospholipids are not polymers (B); four rings describes steroids (C).

6. (C) A peptide bond links the carboxyl carbon of one amino acid to the amino nitrogen of the next, releasing water.

Topic 1.5

1. (B) Backbone hydrogen bonds (C=O ⋯ H–N) form α helices and β sheets, which make up secondary structure.

2. (C) A large pH change alters the charges on R groups, disrupting the ionic and hydrogen bonds that maintain the shape of the active site. Denaturation does not break peptide bonds (A) or change the sequence (B).

3. (A) A substitution that conserves size and polarity, on the surface, is unlikely to disrupt folding.

4. (D) Human enzymes such as amylase are specific for α-glycosidic linkages and cannot hydrolyze the β linkages in cellulose. Cellulose contains no nitrogen (A), starch is a carbohydrate (B), and cellulose molecules are very large (C).

5. (B) Kinked, unsaturated tails prevent tight packing, keeping membranes fluid at cold temperatures.

6. (D) Quaternary structure is the association of multiple polypeptide subunits.

Topic 1.6

1. (A) G = 35%, so C = 35%; G + C = 70%, leaving 30% for A + T, so T = 15%.

2. (A) RNA uses uracil and ribose, which has a 2′ –OH. Thymine and deoxyribose (B) are DNA features; (C) and (D) are shared by both.

3. (C) Pair each base (G–C, A–T): the complement reads 3′–CTAAGT–5′. Rewritten 5′ to 3′, it is 5′–TGAATC–3′. (A) has the right bases but the wrong orientation; (D) uses uracil, which is not in DNA.

4. (C) G–C pairs have three hydrogen bonds versus two for A–T, so more energy is needed to separate G–C-rich DNA.

5. (B) Pairing a double-ring purine with a single-ring pyrimidine keeps every base pair the same width.

Answers: Practice Test

Section I: Answer Key

1 C	2 A	3 D	4 B	5 A	6 C	7 B	8 D	9 A
10 C	11 B	12 A	13 D	14 C	15 B	16 D	17 A	18 C
19 B	20 C	21 A	22 D	23 B	24 D	25 A		

Section I: Explanations

1. (C) Hydrogen bonds form between molecules. Polar covalent bonds (A) hold atoms together *within* one molecule. (Topic 1.1)

2. (A) Cohesive hydrogen bonding at the surface creates surface tension strong enough to support small insects. (1.1)

3. (D) Nitrogen is in the amino group of every amino acid and in every nitrogenous base. (1.2)

4. (B) A tripeptide has two peptide bonds; each requires one water to hydrolyze. (1.3)

5. (A) A phospholipid has a hydrophilic phosphate head and hydrophobic fatty acid tails. (1.4)

6. (C) Phosphorus plus nitrogen plus a pentose sugar identifies a nucleotide. Amino acids lack P; fatty acids and monosaccharides lack N and P. (1.2, 1.4)

7. (B) Cysteine has a sulfhydryl (–SH) group; two cysteines can form an S–S covalent bond. (1.2, 1.5)

8. (D) Chitin, like cellulose, is a structural polysaccharide. The others are energy molecules. (1.4)

9. (A) Enzymes release glucose from chain ends; more branches mean more ends and faster mobilization. (1.5)

10. (C) Hydrophobic interactions drive nonpolar R groups to the interior, away from water. (1.4, 1.5)

11. (B) Replacing a negative charge with a positive one turns an attraction into a repulsion with the positive region of the substrate, weakening binding. (1.5)

12. (A) This is the sugar difference. (B) is reversed; (C) and (D) are false because both have phosphates and backbones. (1.6)

13. (D) C = 18% so G = 18%; G + C = 36%; A + T = 64%; A = 32%. (1.6)

14. (C) Nucleotides are added to the free 3′ –OH, so strands grow 5′ → 3′. (1.4, 1.6)

15. (B) Antiparallel refers to opposite orientation. Bases pair by hydrogen bonds, not covalent bonds (D). (1.6)

16. (D) Breaking hydrogen bonds to evaporate water absorbs a lot of heat from the body. (A) is false: water's specific heat is high. (1.1)

17. (A) Denaturation disrupts weak interactions but not covalent peptide bonds, so the amino acid sequence remains. (1.5)

18. (C) Straight β-linked chains line up so that many –OH groups hydrogen bond between chains, forming strong microfibrils. (1.5)

19. (B) Three ester bonds (glycerol to each of three fatty acids) means three waters released. (1.3)

20. (C) Without double bonds, chains are straight and pack closely. (A) describes unsaturated fats. (1.4)

21. (A) Polar –OH groups hydrogen bond with water; the nonpolar tail cannot. (1.1)

22. (D) Carbon fixed in photosynthesis becomes sugars, which supply carbon skeletons for all other organic molecules. (1.2)

23. (B) The nonpolar valine creates a hydrophobic surface patch; hemoglobin molecules aggregate into fibers, deforming cells. The polypeptide is not shortened (A). (1.5)

24. (D) T_m rises about 4°C for every 10% increase in G–C content. 55% lies halfway between 50% (91°C) and 60% (95°C), so the best estimate is 93°C. (1.6, data analysis)

25. (A) Nonpolar steroids dissolve in the hydrophobic core of the membrane; polar glucose cannot. (1.4)

Section II: Scoring Guidelines

Question 1 (9 points)

(a) 2 points

- 1 pt — Describes the difference: saturated fatty acids have no C=C double bonds (maximum hydrogens); unsaturated fatty acids have one or more C=C double bonds.
- 1 pt — Explains packing: double bonds create kinks that prevent tight packing, increasing membrane fluidity; straight saturated tails pack tightly, decreasing fluidity.

(b) 2 points

- 1 pt — Independent variable: growth temperature. Dependent variable: percentage of unsaturated fatty acids in membrane phospholipids. (Both required.)
- 1 pt — Any one controlled variable: the bacterial species/genetically identical populations, the nutrient medium, growth time (24 hours), number of cultures, extraction method.

(c) 2 points

- 1 pt — Correct axes and labels: x-axis = growth temperature (°C); y-axis = unsaturated fatty acids (%); appropriate scale; points plotted correctly (bar or line graph acceptable because temperature is continuous; a line or scatter plot is preferred).
- 1 pt — Error bars plotted correctly (e.g., 68 ± 3 spans 65–71).

(d) 1 point — Describes an inverse (negative) relationship: as growth temperature increases, the percentage of unsaturated fatty acids decreases. Predicts a value of about 51–52% at 25°C (accept 49–54%).

(e) 2 points

- 1 pt — Explains the benefit: at low temperatures, membranes tend to become rigid; more kinked, unsaturated tails keep the membrane fluid so that membrane proteins can function and substances can move across it.
- 1 pt — Predicts that the mutant would grow more slowly (or not at all) at 10°C, with justification: with only saturated tails, its membranes would pack tightly and become too rigid at low temperature, impairing transport and other membrane functions.

Question 2 (4 points)

- **(a) 1 pt** — Describes any two: hydrophobic interactions among nonpolar R groups; hydrogen bonds between polar R groups; ionic bonds between oppositely charged R groups; covalent disulfide bridges between cysteines.
- **(b) 1 pt** — Predicts that enzyme activity will decrease or be lost.
- **(c) 1 pt** — Justifies: a charged R group cannot participate in hydrophobic interactions in the nonpolar core and is energetically unfavorable there; the protein misfolds, altering tertiary structure and the shape of the active site, even though the mutation is far from the active site.
- **(d) 1 pt** — Explains that isoleucine is also nonpolar and of similar size, so it can still participate in the hydrophobic interactions in the core; the overall fold, and thus the active site, is largely preserved.

Question 3 (4 points)

- **(a) 1 pt** — 20% (100 − 30 − 30 − 20 = 20; also, %C = %G in double-stranded DNA).

- **(b) 2 pts** — 1 pt: identifies Sample 2. 1 pt: justifies that %A ≠ %T (24 vs. 32) and %G ≠ %C (18 vs. 26), so the bases are not complementary-paired; Chargaff's rules apply only to double-stranded DNA. (The sample contains thymine, so it is DNA rather than RNA.)
- **(c) 1 pt** — Sample 3, because it has a higher G–C content (64% vs. 40%), and G–C pairs are held by three hydrogen bonds compared with two for A–T pairs, so more energy is needed to separate the strands.

INTERPRETING YOUR SCORE

Multiple choice: 22–25 correct = excellent command of Unit 1; 17–21 = solid, review missed topics; below 17 = reread the lessons for the topics you missed and redo their Check Your Understanding sets. **Free response:** 14–17 points = strong; 10–13 = good, focus on complete "because" explanations; below 10 = practice the task verbs in Part 2. These ranges are study guidance only and do not predict an official AP score.

Glossary

Adhesion
Attraction between molecules of different substances, such as water and glass.

Amino acid
Monomer of proteins: amino group, carboxyl group, hydrogen, and R group on a central carbon.

Amphipathic
Having both hydrophilic and hydrophobic regions.

Antiparallel
Arrangement of DNA strands running in opposite 5′→3′ directions.

Capillary action
Rise of water in a narrow tube due to adhesion and cohesion.

Carbohydrate
Sugar or sugar polymer made of C, H, and O.

Cellulose
Structural polysaccharide of β-glucose in plant cell walls.

Chargaff's rules
In double-stranded DNA, %A = %T and %G = %C.

Chitin
Structural polysaccharide in fungal cell walls and arthropod exoskeletons.

Cholesterol
A steroid in animal membranes; precursor to steroid hormones.

Cohesion
Attraction between molecules of the same substance.

Complementary base pairing
A with T (or U) and G with C via hydrogen bonds.

Covalent bond
A strong bond formed by sharing electrons.

Dehydration synthesis
Reaction joining monomers by removing water.

Denaturation
Loss of a protein's native shape due to heat, pH, or chemicals.

Deoxyribose
Five-carbon sugar of DNA; lacks an –OH at the 2′ carbon.

Disulfide bridge
Covalent bond between sulfur atoms of two cysteines.

Electronegativity
An atom's pull on shared electrons.

Enzyme
A biological catalyst, usually a protein.

Evaporative cooling
Cooling of a surface as high-energy water molecules evaporate.

Fatty acid
Hydrocarbon chain with a carboxyl group at one end.

Functional group
Atom cluster that gives a molecule characteristic properties.

Glycerol
Three-carbon alcohol that forms the backbone of fats and phospholipids.

Glycogen
Highly branched storage polysaccharide in animals.

Glycosidic linkage
Covalent bond between two monosaccharides.

Hydrogen bond
Weak attraction between a δ+ hydrogen and a δ− O or N atom.

Hydrolysis
Breaking a bond by adding water.

Hydrophilic
Interacting readily with water; polar or charged.

Hydrophobic
Not interacting favorably with water; nonpolar.

Hydrophobic interaction
Clustering of nonpolar groups away from water.

Ionic bond
Attraction between oppositely charged ions.

Isomers
Compounds with the same formula but different structures.

Lipid
Diverse hydrophobic molecules: fats, phospholipids, steroids.

Monomer
Small building-block molecule of a polymer.

Monosaccharide
Simple sugar such as glucose or fructose.

Nucleic acid
Polymer of nucleotides (DNA or RNA).

Nucleotide
Pentose sugar + phosphate group + nitrogenous base.

Peptide bond
Covalent bond joining two amino acids.

pH
$-\log[H^+]$; measures acidity on a 0–14 scale.

Phosphodiester bond
Bond linking nucleotides through a phosphate group.

Phospholipid
Glycerol + two fatty acids + phosphate head; forms membranes.

Polar molecule
Molecule with an uneven distribution of charge.

Polymer
Large molecule made of many linked monomers.

Polypeptide
A chain of amino acids linked by peptide bonds.

Polysaccharide
A polymer of many monosaccharides.

Primary structure
Amino acid sequence of a polypeptide.

Protein
One or more folded polypeptides with a specific function.

Purine
Double-ring nitrogenous base: A, G.

Pyrimidine
Single-ring nitrogenous base: C, T, U.

Quaternary structure
Association of two or more polypeptide subunits.

R group
Variable side chain of an amino acid.

Ribose
Five-carbon sugar of RNA with a 2′ –OH.

Saturated fatty acid
Fatty acid with no C=C double bonds.

Secondary structure
α helices and β pleated sheets from backbone hydrogen bonds.

Specific heat
Energy to raise 1 g of a substance by 1°C.

Starch
Storage polysaccharide of α-glucose in plants.

Steroid
Lipid with four fused carbon rings.

Surface tension
Resistance of a liquid surface to breaking, from cohesion.

Tertiary structure
Overall 3-D shape of a polypeptide from R-group interactions.

Triglyceride
Glycerol bonded to three fatty acids; a fat.

Unsaturated fatty acid
Fatty acid with one or more C=C double bonds.

Score Tracker

Record each attempt. Redo any set where you scored below 80% after reviewing the lesson.

PRACTICE SET	ATTEMPT 1	ATTEMPT 2	ATTEMPT 3	TOPICS TO REVIEW
Check Your Understanding 1.1 (/6)				
Check Your Understanding 1.2 (/5)				
Check Your Understanding 1.3 (/5)				
Check Your Understanding 1.4 (/6)				
Check Your Understanding 1.5 (/6)				
Check Your Understanding 1.6 (/5)				
Practice Test — Section I (/25)				
Practice Test — FRQ 1 (/9)				
Practice Test — FRQ 2 (/4)				
Practice Test — FRQ 3 (/4)				

Learning Target Checklist

Check each statement only when you can do it without notes.

☐ I can explain why water is polar and draw hydrogen bonds between water molecules.

☐ I can link cohesion, adhesion, specific heat, heat of vaporization, ice density, and solvency to hydrogen bonding, with an example of each.

☐ I can state which macromolecules contain N, P, and S.

☐ I can draw and label dehydration synthesis and hydrolysis.

☐ I can draw an amino acid and a nucleotide and identify their parts.

☐ I can compare saturated and unsaturated fatty acids and explain phospholipid bilayers.

☐ I can describe all four levels of protein structure and the interactions that stabilize them.

☐ I can predict and justify the effect of an amino acid substitution or denaturing condition.

☐ I can explain why starch is digestible and cellulose is structural.

☐ I can compare DNA and RNA and write a complementary strand with 5′ and 3′ ends labeled.

☐ I can use Chargaff's rules to calculate base percentages.